

KCPC-CS120

Supplementary Reading

Combinatorics I: Learning to Count

This document contains recommended and further readings for stars and bars, principle of inclusion-exclusion, bijections, and Burnside's Lemma (which requires knowledge of group theory).

These references are intended to deepen conceptual understanding beyond implementation.

I. Core Algorithmic Reading

Competitive Programmer's Handbook — Antti Laaksonen (PDF)

<https://cses.fi/book/book.pdf>

- **Counting** — Chapter 22, p. 227
- **Binomial Coefficients** — Section 22.2, p. 229
- **Inclusion-Exclusion** — Section 22.4, p. 232
- **Burnside's Lemma** — Section 22.6, p. 235
- **Catalan Numbers** — Section 22.7, p. 236

II. Online Reference — CP-Algorithms

Combinatorics

<https://cp-algorithms.com/combinatorics/>

- **Binomial Coefficients** —
<https://cp-algorithms.com/combinatorics/binomial-coefficients.html>
 - **Stars and Bars** —
https://cp-algorithms.com/combinatorics/stars_and_bars.html
 - **Inclusion-Exclusion** —
<https://cp-algorithms.com/combinatorics/inclusion-exclusion.html>
 - **Catalan Numbers** —
<https://cp-algorithms.com/combinatorics/catalan-numbers.html>
 - **Burnside's Lemma** —
<https://cp-algorithms.com/combinatorics/burnside.html>
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III. Formal Textbook Reference

Dexter Chua's Lecture Notes

These are intended for a more advanced audience comfortable with proof-based mathematics or those willing to learn more about group theory or number theory.

Numbers and Sets: Covers bijections, counting arguments, inclusion-exclusion, and the formal foundations of combinatorics.

https://dec41.user.srcf.net/h/IA_M/numbers_and_sets

Groups: Required background for Burnside's Lemma. Read this only if you want to understand the group-theoretic proof rather than just applying the formula.

https://dec41.user.srcf.net/h/IA_M/groups

The ArtofProblemSolving has a lot of material on all sorts of olympiad content, recommended to check it out. It's intended more for people who've already done some olympiads though.

Brilliant is good for newbies and explains everything from the ground up.